Rice - Brown spot

RAG record ADV-006 | canonical key: rice_brown_spot

Record ID	ADV-006
Crop / crop key	Rice / rice
Condition / disease key	Brown spot / brown_spot
Main pathogen	Bipolaris oryzae (syn. Helminthosporium oryzae)
RAG aliases	rice brown spot, Bipolaris oryzae, Helminthosporium oryzae, sesame leaf spot

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Typical lesions are round to oval brown spots, often with a lighter or grey centre, scattered over the leaf.
- Severe disease can cause seedling blight, larger dead patches and poor grain quality.

Weather or field conditions that favour it

Nutrient-poor or water-stressed crops are more vulnerable. Infected seed can also carry the fungus.

What to do now

- Correct plant stress first: maintain good irrigation and balanced soil fertility.
- Use disease-free seed and treat seed before sowing where brown spot is a known problem.
- If disease is active in the field, use a registered fungicide and cover the foliage well.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Captan or thiram seed treatment	2 g/kg seed	TNAU seed treatment recommendation for reducing primary infection.	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
Mancozeb	2 g/L water	TNAU foliar option; 2-3 sprays at 10-15 day intervals may be used when disease requires treatment.	[S1]
Edifenphos	1 ml/L water	Alternative foliar option listed by TNAU.	[S1]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Use clean seed.
- Maintain balanced nutrition, especially in poor soils.
- Avoid water stress.
- Use resistant/tolerant varieties where recommended.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Tamil Nadu Agricultural University (TNAU). Rice - Brown Spot management.
https://www.agritech.tnau.ac.in/expert_system/paddy/cpdisbrownspot.html